

CSC311 Introduction to Machine Learning Week 12 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W12-T1. Explain why dimensionality reduction is an interesting unsupervised learning task. What can we do with low-dimensional data that we cannot do with high-dimensional data?

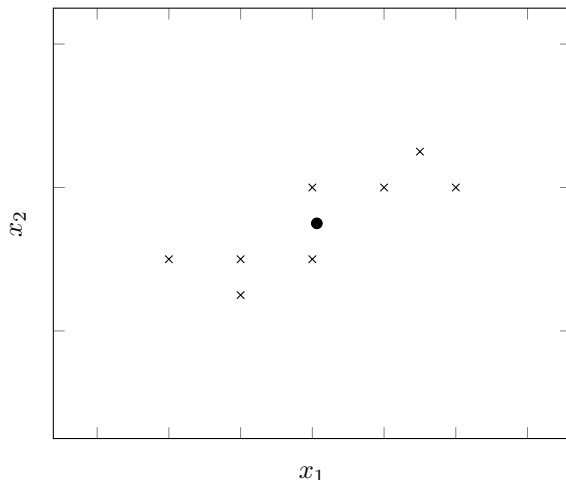
W12-T2. PCA is a *linear* dimensionality reduction technique. Explain what that means.

W12-T3. Project the vector $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ on to the subspace spanned by $\mathbf{u}_1 = \begin{bmatrix} -\sqrt{2} \\ \sqrt{2} \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

W12-T4. For this question, we will work with the data $\mathbf{X} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & -2 & 2 \end{bmatrix}$, and the basis vectors $\mathbf{u}_1 = \begin{bmatrix} -\sqrt{2} \\ \sqrt{2} \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

- (a) Compute $\hat{\boldsymbol{\mu}}$, the empirical mean of the data.
- (b) Compute the representations $\mathbf{Z}$ when $\mathbf{X}$ is projected onto the space centered at $\hat{\boldsymbol{\mu}}$ and spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$. (Notation question: how is $\mathbf{Z}$ related to each $\mathbf{z}^{(i)}$)?
- (c) Compute the reconstruction $\tilde{\mathbf{x}}^{(1)}$ and $\tilde{\mathbf{x}}^{(2)}$.
- (d) Compute the reconstruction error.
- (e) Compute the variance of the reconstruction.
- (f) Compute the variance of the representations.
- (g) Explain why the previous two quantities are equivalent.

W12-T5. In this question, we would like to use PCA to reduce the dimensionality of this data set in $\mathbb{R}^2$, marked with “x” below. The center of the data is marked with a solid “o”. Suppose that PCA is performed on the data below. Sketch the two principal component directions in the plot below. Label the first and second PCA directions $\mathbf{u}_1$ and $\mathbf{u}_2$.



W12-T6. Suppose that we are performing PCA on a data set $\mathbf{x}^{(i)} \in \mathbb{R}^D$. We use $\hat{\boldsymbol{\mu}}$ to represent the empirical mean of $\mathbf{x}^{(i)}$, and choose a matrix $\mathbf{U}$ of shape $D \times K$ with orthonormal columns, representing the basis of our PCA space. so, we have the projected features $\mathbf{z}^{(i)} = \mathbf{U}^T(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})$, and the reconstruction $\tilde{\mathbf{x}}^{(i)} = \hat{\boldsymbol{\mu}} + \mathbf{U}\mathbf{z}^{(i)}$. Prove that $\tilde{\mathbf{x}}^{(i)} - \hat{\boldsymbol{\mu}}$ is orthogonal to $\tilde{\mathbf{x}}^{(i)} - \mathbf{x}^{(i)}$.

Validation

W12-V1. Suppose we have a data set $\mathcal{D} = \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)}\}$, where $\mathbf{x}^{(i)} \in \mathbb{R}^D$. In Principal Component Analysis, we reduce the dimensionality of this data by mapping each data point $\mathbf{x}^{(i)}$ to a lower dimensional representation $\mathbf{z}^{(i)} \in \mathbb{R}^K$:

$$\mathbf{z}^{(i)} = \mathbf{U}^T(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}),$$

where $\hat{\boldsymbol{\mu}}$ is the empirical mean of the data set. Show that the mean of the codes, $\frac{1}{N} \sum_{i=1}^N \mathbf{z}^{(i)}$, is the zero vector.

W12-V2. On the data set from question W12-T5, Lisa tried to compute the empirical covariance matrix of the data above and obtained

$$\hat{\boldsymbol{\Sigma}} = \begin{bmatrix} 13.2 & -0.5 \\ -0.5 & 3.5 \end{bmatrix}.$$

She immediately suspects that his calculation was incorrect. Without performing any mathematical computations, explain why she might have this suspicion.

Generalization

W12-G1. No generalization questions this week. Please ask about autoencoders in office hours if you are interested in other dimensionality reduction techniques!